Curriculum Vitae | Adam Thomas Clare, MEng, PhD, FHEA

Nationality | British **Email** | adam.clare@nottingham.ac.uk **Phone** | 00 44 7821988194

Professional Summary

● Passion for communication of science and delivering impact at scale.● Professor of Manufacturing Engineering with extensive experience of fundamental and industrially relevant research having created and **transitioned multiple technologies from the lab to industrial proposition** ● Demonstrated academic leadership currently heading an international and diverse research team of 25 research staff and students that has activities across additive manufacturing/3D printing and high value manufacturing ● Active multi-year strategic partnerships held with Rolls-Royce (UK), Oerlikon (USA/Germany) and Mitsubishi (Japan) ● Extensive experience of working and collaborating in Asia, Europe and North America ● Proven leadership (selecting, resourcing and delivering) as Director of Admissions and Chair of Research Operations Group creating policy and business planning for an annual UG intake of ~700 students and PG intake of 400-600 students (**annual revenue to business of ~£10M**) ● Demonstrated leadership to staff both within and outside my line management

Employment History

Research Chair | Rolls-Royce Royal Academy Research Chair	2019 – Present
Founder, Non-Executive Director & Scientific Advisor |Texture Jet Ltd.	2019 – Present
Secretary | CIRP STC-E	2019 – Present
Professor | Kanazawa University, Japan	2019 - Present
Deputy Director, Rolls-Royce UTC | University of Nottingham	2018 - Present
Li Day Sum Professor of Advanced Manufacturing | University of Nottingham, Ningbo, China	2018 - Present
Full Professor of Manufacturing Engineering | University of Nottingham	2018 - Present
Secondment | Rolls-Royce PLC Submarines	2016- 2017
Associate Professor of Advanced Manufacturing | University of Nottingham	2015 - 2017
Assistant Professor of Advanced Manufacturing | University of Nottingham	2010 - 2015
Research Associate | University of Liverpool	2008 - 2009

Academic Credentials

PhD, "Layer Based Manufacturing of Functional Materials and Components" | 2009
MEng (Hons), Integrated Engineering | 2005
University of Liverpool

Certificate in Professional Studies "Teaching and Learning in Higher Education" | 2009

Teaching Commitments

Undergraduate

"Group Design and Make" (MM4GDM, Convenor)

- Instruct 100-170 students in a mixture of both small group and lecture based format.
- BEng and MEng Dissertation Projects (6 Students per Annum)

Postgraduate

- Additive Manufacturing - Group Grand Challenge | A group based module exploring AM technologies and their incorporation into underpinning research.
- BEng and MEng Dissertation Projects (6 Students per Annum)

PhD Supervision

- Currently advise 17 students (7 industry-funded); 15 PhDs completed
- External Review of PhD students | Birmingham, Cranfield, Sheffield, Cambridge, KU Leuven, Basque Country, EPFL, Cardiff, NTU Singapore

Internet Based Resources

Click each icon to launch the respective page.

Research Funding Overview

Achieved over £6M in Research Grants as PI, and in excess of £13.5M total grants since appointment at the U of Nottingham.

Dates	Funding Body & Title	Role	Valuation
2020 - 2023	InNovemberate UK, ATI, Rolls-Royce, "RESINSTATE"	CI	£1,900,000
2020 - 2024	Rolls-Royce PhD Sponsorship | Hendrik Groth	PI	£67,000
2020 - 2024	Siemens Turbomachinery PhD Sponsorship | Thomas Girerd	PI	£82,000
2019 - 2021	Mitsubishi Electric Europe: Direct Support, Provision of Machinery, EDM Coating Apparatus	PI	£167,000
2019 - 2024	Royal Academy of Engineering/ Rolls-Royce Research Chair	PI	£850,000
2019 - 2024	EPSRC Program Grant, UK FIRES EP/S019111/1	PI/CI	£410,000
2018 - 2019	Various Confidential Projects, Rolls-Royce	PI	£320,000
2018 - 2019	IAA Research Grant | Jonathon Mitchell-Smith PDR	PI	£56,000
2018 - 2022	Oerlikon Balzers, PhD Studentship | Timothy Cooper	PI	£62,000
2018 - 2019	Fastener Removal, Rolls-Royce	PI	£156,000
2018 - 2020	EPSRC - UK Research Centre In Non-Destructive Evaluation – SRAS for AM	PI	£194,000
2018 - 2019	IAA – EJM Advancement	PI	£62,000
2017 - 2020	Rolls-Royce PhD Studentship | Nuhaize Ahmed	PI	£66,000
2017 - 2020	Oerlikon Metco PhD Studentship | Alex Gullane	PI	£80,000
2016 - 2019	Oerlikon Metco PhD Studentship| Alexander Gasper	PI	£30,000
2016 - 2018	EPSRC (EP/N034201/1) High Resolution Biomedical Imaging Using Ultrasonic Metamaterials	PI	£202,000
2015 - 2018	Siemens PhD Studentship | Sam Catchpole-Smith	PI	£80,000
2014 - 2022	EPSRC (EP/L01534X/1) Centre for Doctoral Training in Additive Manufacturing and 3D Printing	CI	£4,557,035
2016 - 2019	InNovemberate UK, Functional Lattices for Automotive Components (FLAC)	CI	£1,731,610
2014 - 2020	EPSRC (EP/L022125/1): UK Research Centre In Non-Destructive Evaluation – Laser Acoustic Emission for Integrity Monitoring in SLM	CI	£600,000
2014 - 2016	EPSRC (EP/L01713X/1): In-situ Monitoring of Component Integrity During Additive Manufacturing Using Optical Coherence Tomography	PI	£144,323

2014 - 2016	EPSRC (EP/L017482/1): Functional Surfaces via Non-Conventional Processes	PI	£246,961
2013 - 2019	Mitsubishi Electric Europe: Direct Support, Provision of Machinery, EDM Coating Apparatus	PI	£212,000
2013 - 2016	EPSRC (EP/K034987/1): Underpinning Power Electronics - Integrated Drives	CI	£489,598
2013 - 2016	TSB/Renishaw/Delcam/ Alcon: ALSAM (Lightweight Aluminum Structures for Automotive Applications)	CI	£229,586
2013 - 2014	Royal Academy of Engineering: Industrial Secondment – Rolls-Royce SubMarchines	PI	£26,000
2012 - 2017	Cummins Generator Technologies: 2 x EngD studentships	PI/CI	£140,000

Research Funding Overview, Continued

2012 - 2014	BAE Systems: Advanced Tooling for Waterjet Masking	CI	£96,000
2012 - 2014	JSPS: Bilateral Research Partnership – "Micro-EDM of Single Crystal Silicon" with University of Tokyo	CI	£30,000
2013	JSPS: Furusato Award (Received at Japanese Embassy)	PI	£2,000
2011 - 2012	JSPS: Invited Fellow - University of Tokyo	PI	£15,000
2010	JSPS: Short Term Fellow - University of Tokyo	PI	£10,000
2012	Nuffield Foundation: Process Analysis of Electrical Discharge Machining – Undergraduate Summer Placement	PI	£1,500
2012	EPSRC (µProject in EP/I016813/1): Characterisation of Aerospace Materials Processed by Electrical Discharge Machining and Electron Beam Melting	PI	£24,000
2010 - 2012	Transport iNET: Improved Longevity of Railway Track Using Laser Cladding	PI	£137,691

Honours, Distinctions & Awards

2X ISEM Best Paper Award, CIRP ISEM	2018
Excellent Paper Award, CIRP ISEM *"Electrolyte Jet Machining of Titanium Alloys Using Novel Electrolytes"*	2016
Visiting Academic, Sichuan University, China	2015- 2017
Presenter, Universities of Tokyo, Okayama, Kyoto, Warwick and University College Dublin	2011-2016
FAME Award Recipient, Solid Freeform Fabrication Conference (Austin, TX) for Contributions to Additive Manufacturing and 3D printing	2014
Furusato Award Recipient, Collaboration with University of Tokyo	2012
Invited Fellow, Japan Society for the Promotion of Science at the University of Tokyo	2011

Research Related Activities

Knowledge Exchange

Industrial partners include: Transport iNet, Rolls Royce, Renishaw (MTT), Mitsubishi Electric Europe/Japan, Cummins Generator Technologies, Siemens, Oerlikon Balzers/Metco, Oerlikon AM, TADA Electric. Scientific Advisor and Director at Texture Jet Ltd – University Spin out company.

Inventions & Patents

"Satelliting for Additive Manufacturing Materials", Patent Filed September 2013

Further 4 disclosures to University of Nottingham IP Office:

- Salt Blend Preparations for SLM
- Electrolyte Development for EJM
- Nozzle Arrangement and Operation for EJM
- µwave Detect and Repair Technologies

4 filed patents with Rolls-Royce Plc 'An Electro-Discharge Machining Tool And A Method Of Using The Same' 2018
2 Further Disclosures Made with Rolls-Royce

Public Outreach
Nottingham Lead – '88 Pianists' Outreach Program
Local School Host, University of Nottingham STEM Outreach Programme
Annual Presenter, Reading School for Boys

Additional Activities & Engagements

External Examiner | Warwick University	2020-Present
SLPC April 21-23, 2020, Yokohama, Japan	2020-Present
Scientific Advisory Board | ISEM, Switzerland	2020-Present
Scientific Advisory Board | SFI Manufacturing Centre, Ireland	2020-Present
Scientific Advisory Board | EPSRC Future Formulations	2018-Present
Faculty of Engineering Director of Admissions | University of Nottingham	2018-Present
Member | EPSRC Strategic Advisory Team (SAT), Manufacturing the Future Theme	2018-Present
External Examiner | Queen's University Belfast	2018-Present
Li Dak Sum Chair of Additive Manufacturing, University of Nottingham Ningbo Campus	2018-Present
Board of Reviewers | International Journal of Machine tools and Manufacturing	2017-Present
Associate Editor | Journal of Materials Processing Technology	2017-Present
Subject Editor | Precision Engineering	2017-Present
Founding Member | EPSRC Future Leaders Initiative in Engineering	2016-Present
Member | EPSRC College of Reviewers	2016-Present
Member | EPSRC College of Reviewers	2015-Present
Reviewer | Deutsche Forschungsgemeinschaft (German Research Foundation) Fonds de recherche du Québec – Nature et technologies, Knowledge Foundation (Sweden)	2012-Present
Chairman | Faculty of Engineering Research Operations Group	2014-2016
Founding Member | EPSRC Early Career Forum in Manufacturing	2013-2016
Rolls-Royce Nuclear, Royal Academy of Engineering Secondment	2013-2014

Publication List

Summary

- Over 134 Journal Publications in high quality peer reviewed international journals across manufacturing technology and materials science.
- Average of 10+ journal publications yearly in high quality journals since appointment at Nottingham in 2010
- Currently have 6 papers ranked in the top 10 most downloaded across journals (Materials & Design, Journal Materials Processing Technology and Surface & Coatings Technology. Including the most downloaded in Materials & Design).

Majority of work is published in journals relating to manufacturing processes and material science, but more recently has expanded into general science including scientific reports and nature communications. Since taking up an academic position, have been listed as the corresponding author and not the first author in most cases, a common practice to develop the exposure of students involved in the work. Where listed as corresponding author or last author my contribution is in the conception, planning, resource allocation, oversight of the experimental work and production of the paper.

Peer Reviewed Journal Publications (Hyperlinked)

1) G.J. Chaplain, J. M. De Ponti, A. Colombi, R. Fuentes-Dominguez, P. Dryburg, D. Pieris, R.J. Smith, A.T. Clare, M. Clark And R.V. Craster, (2020). Tailored Elastic Surface to Body Wave Umklapp Conversion Nature Communications.

2) Stefano Laureti, D.A. Hutchins, L. Astolfi, R.L. Watson, P.J. Thomas, P. Burrascano, L. Nie, S. Freear, M. Askari, A.T. Clare And M. Ricci, (2020). Trapped Air Metamaterial Concept for Ultrasonic Sub-Wavelength Imaging In Water Scientific Reports

3) A.T. Clare*, W.J. Reynolds, J.W. Murray, N.T. Aboulkhair, M. Simonelli, M. Hardy, D.M. Grant And C. Tuck, (2020). Laser Calorimetry For Assessment Of Melting Behaviour In Multi-Walled Carbon Nanotube Decorated Aluminium By Laser Powder Bed Fusion Cirp Annals

4) Speidel, Alistair; Murray, James; Bisterov, Ivan; Mitchell-Smith, Jonathon; Parmenter, Christopher; Clare, Adam* (2020). Thermal Activation of Electrochemical Seed Surfaces for Selective and Tuneable Hydrophobic Patterning ACS Applied Materials and Interfaces

5) R.Sebastian, H. Chen, A. Rushworth, A.T Clare*(2020) Unit Cell Type Scan Strategies For Powder Bed Fusion: The Hilbert Fractal Additive Manufacturing.

6) Hutchins, D, M. Askari, R. Watson, L. Nie, R.Wildman, L. Astolfi, S. Freear. P. Thomas, S. Laureti, M. Clark, E. Woods, C. Tuck, A.T. Clare* (2020) Additive Manufacturing of Metamaterials: A Review Additive Manufacturing

7) P. Smith, J. Murray, J. Segal, A.T. Clare* (2020) Magnetically Assisted Directed Energy Deposition Journal of Materials Processing Technology.

8) Murray, James W., Ahmed, Nuhaize, Yuzawa, Takashi, Nakagawa, Takayuki, Sarugaku, Shun, Saito, Daiki And Clare, Adam T*.(2020) Dry-Sliding Wear and Hardness of Thick Electrical Discharge Coatings and Laser Clads Tribology International.

9) Ahmed, N., Murray, J. W., Yuzawa, T., Nakagawa, T., Sarugaku, S., Saito, D., Brown, P. D. And Clare, A. T.*, (2020). Formation of Thick Electrical Discharge Coatings: Journal Of Materials Processing Technology.,

10) D. Hutchins, M. Askari, R. Watson, L. Nie, A. Lorenzo, S. Freear, S. Laureti, M. Clark And A. Clare*, (2020). An Ultrasonic Metallic Fabry-Pérot Metamaterial for Use in Water Additive Manufacturing.

11) Gorji, N. E., Saxena, P., Corfield, M., Clare, A., Rueff, J. P., Bogan, J., González, P. G. M., Snelgrove, M., Hughes, G., O'connor, R., Raghavendra, R. And Brabazon, D. (2020). A New Method For Assessing The Utility Of Powder Bed Fusion (Pbf) Feedstock: Materials Characterization Materials Characterization

12) N. Aboulkhair, R. Hague, A. Clare*, A. Kennedy, G. Zhao And I. Ashcroft, (2020) Generation Of Graded Porous Structures By Control Of Process Parameters In The Selective Laser Melting Of A Fixed Ratio Salt-Metal Feedstock Journal Of Manufacturing Processes.

13) Patrick Bointon, Luke Todhunter, Adam Clare, Richard Leach (2020) Performance Verification Of A Flexible Vibration Monitoring System Machines

14) Mohaimen Al-Thamir, D. Graham Mccartney,Marco Simonelli,Richard Hague And Adam Clare* (2020) Processability Of Atypical Wc-Co Composite Feedstock By Laser Powder-Bed Fusion Materials

15) Marco Simonelli, Graham McCartney, Pere Barriobero-Vila, Nesma Aboulkhair, Yau Yau Tse, Adam Clare, Richard Hague (2020) Controlling Recrystallization of Ti-6al-4v Using Additions of Fe In Pbf Metallurgical and Materials Transactions

16) Long Bai, Alexander Velichko, Adam Clare, Paul Dryburgh, Don Pieris, Bruce Drinkwater (2020) The Effect of Distortion Models on Characterisation Of Real Defects Using Ultrasonic Arrays NDT& E International

17) Olusola Oyelola, Alexander Jackson-Crisp, Pete Crawforth, Don M. Pieris, Richard J. Smith, Rachid M'saoubi, Adam T. Clare* (2020). Machining of Directed Energy Deposited Ti6al4v Using Adaptive Control Journal of Manufacturing Processes

18) T.R. Walker, C.J .Bennett, T.L.Lee, A.T.Clare (2020). A Novel Numerical Method to Predict the Transient Track Geometry and Thermomechanical Effects Through In-Situ Modification of the Process Parameters in Direct Energy Deposition Finite Elements in Analysis and Design

19) Don Pieris, Theodosia Stratoudaki, Yashar Javadi, Peter Lukacs, Sam Catchpole-Smith, Paul D.Wilcox, Adam T. Clare And Matt Clark, (2019). Laser Induced Phased Arrays (LIPA) to Detect Nested Features in Additively Manufactured Components Materials & Design.

20) K. Al-Hamdani, J. Murray, T. Hussain and A.T Clare*, (2019). Controlling Ceramic-Reinforcement Distribution in Laser Cladding of MMCS Surface & Coatings Technology.

21) Wessel W.Wits, Marc De Smit, Kamaalal-Hamdani And Adam T.Clare, (2019). Laser Powder Bed Fusion of a Magnesium-Sic Metal Matrix Composite Procedia Cirp

22) Pieris, D., Patel, R., Dryburgh, P., Hirsch, M., Li, W., Sharples, S. D., Smith, R. J., Clare, A. T. And Clark, M., (2019). Spatially Resolved Acoustic Spectroscopy Towards Online Inspection of Additive Manufacturing: Insight: Non-Destructive Testing and Condition Monitoring

23) F Zhang, T Hua, A.T. Clare, W. Huang, J. Chen, X. Lin And M. Guo (2019) Microstructure and Properties of Ti-6al-4v Fabricated by Low-Power Pulsed Laser Directed Energy Deposition Journal of Materials Science & Technology.

24) K Al-Hamdany, J.W Murray, T. Hussain And A.T. Clare, (2019). Heat-Treatment and Mechanical Properties of Cold-Sprayed High Strength Al Alloys from Satellite Feedstocks Surface and Coatings Technology.

25) H. Tan, T. Hu, F. Zhang, Y. Qiu And A.T. Clare, (2019). Direct Metal Deposition of Satellite Ti-15mo: Microstructure and Mechanical Properties Advanced Engineering Materials

26) A Gasper, D Hickman, I Ashcroft, S Sharma, X Wang, B Szost, D Johns and A.T. Clare*, (2019) Oxide and Spatter Powder Formation During Laser Powder Bed Fusion of Hastelloy X Powder Technology.

27) A Speidel, J Mitchell-Smith, I Bisterov And A.T. Clare (2019). Oscillatory Behaviour In the Electrochemical Jet Processing of Titanium Journal of Materials Processing Technology.

28) Wits, W. W., De Smit, M., Al-Hamdani, K. And Clare, A. T., (2019) Laser Powder Bed Fusion Of A Magnesium-Sic Metal Matrix Composite Procedia Cirp

29) Farayibi, P. K., Abioye, T. E., Kennedy, A. And Clare, A. T., (2019) Development of Metal Matrix Composites by Direct Energy Deposition Of 'Satellite' Powders Journal of Manufacturing Processes.

30) Catchpole-Smith, S., Sélo, R. R. J., Davis, A. W., Ashcroft, I. A., Tuck, C. J. And Clare, A., (2019) Thermal Conductivity Of TPMS Lattice Structures Manufactured Via Laser Powder Bed Fusion: Additive Manufacturing Additive Manufacturing

31) P. Dryburgh, D. Pieris, F. Martina, R. Patel, W. Li, R. Smith, S. D. Sharples, S. Williams, A.T. Clare (2019) Spatially Resolved Acoustic Spectroscopy for Integrity Assessment in Wire-Arc Additive Manufacturing Additive Manufacturing

32) T.R.Walker, C.J.Bennett, T.L.Lee And A.T.Clare, (2019). A Validated Analytical-Numerical Modelling Strategy to Predict Residual Stresses in Single-Track Laser Deposited In718 International Journal of Mechanical Sciences.

33) J. Walker, J. Mitchell-Smith and A.T. Clare, (2019). Influence of Contact Area on The Sliding Friction and Wear Behaviour Of an Electrochemical Jet Textured Al-Si Alloy Wear

34) A.T. Clare, J. Mitchell-Smith, A. Speidel, I. Bisterov and J.W Murray, (2019). Surface Enhanced Micro Features Using Electrochemical Jet Processing Cirp Annals.

35) J.W Murray, D Grant, M Simonelli, A.T. Clare And A Speidel, (2019). Spheroidisation Of Metal Powder By Pulsed Electron Beam Irradiation Powder Technology

36) S.J. Algodi, J. Murray, P.D. Brown And A.T. Clare, (2018). Wear Performance Of Tic/Fe Cermet Electrical Discharge Coatings Wear.

37) James W Murray, Richard B Cook, Nicola Senin, Samer J Algodi And Adam T Clare, (2018). Defect-Free Tic/Si Multi-Layer Electrical Discharge Coatings Materials And Design

38) Jacob Williams, Paul Dryburgh, Prahalada Rao, Adam Clare And Ashok Samal, (2018). Defect Detection And Monitoring In Metal Additive Manufactured Parts Through Deep Learning Of Spatially Resolved Acoustic Spectroscopy Signals Smart And Sustainable Manufacturing Systems.

39) Z Xu, C Hyde, J Murray And A T Clare, (2018). Effect Of Post Processing On The Creep Performance Of Laser Powder Bed Fused Inconel Additive Manufacturing.

40) A Gasper, I Ashcroft, D Johns, B Szost, S Sharma And A T Clare, (2018). Spatter And Oxide Formation In Laser Powder Bed Fusion Of Inconel 718 Additive Manufacturing.

41) Wang, M., Zhang, Y., Xu, X., Chen, G., Clare, A. T. And Ahmed, N., (2018). Effects Of Tool Intermittent Vibration On Helical Internal Hole Processing In Electrochemical Machining Part C: Journal Of Mechanical Engineering Science Proceedings Of The Institution Of Mechanical Engineers

42) Olusola Oyelola, P Crawforth, R M'saoubi And A.T.Clare, (2018). Machining Of Functionally Graded Ti6al4v/ Wc Produced By Directed Energy Deposition Additive Manufacturing.

43) Oreffo, R., Gorianov, V., Cook, R., Walker, J., Dunlop, D. G., & Clare, A. (2018) Human Skeletal Stem Cell Response To Multiscale Topography Induced By Large Area Electron Beam Melting Surface Treatment. Frontiers In Bioengineering And Biotechnology, 1-12.

44) Alistair Speidel, Jonathon Mitchell-Smith, Ivan Bisterov And Adam Thomas Clare, 2018. The Importance of Microstructure in Electrochemical Jet Processing Journal of Materials Processing Technology.

45) Alistair Speidel, Rong Su, Jonathon Mitchell-Smith, Paul Dryburgh, Ivan Bisterov, Don M Pieris, Wenqi Li, Rikesh Patel, Matt Clark And Adam Thomas Clare, 2018. Crystallographic Texture Can Be Rapidly Determined By Electrochemical Surface Analytics Acta Materialia.

46) Marco Simonelli, Nesma T Aboulkhair, Philip Cohen, James Murray, Adam Clare, Christopher Tuck And Richard Hague, 2018. A Comparison of Ti-6al-4v In-Situ Alloying in Selective Laser Melting Using Simply-Mixed and Satellite Powder Blend Feedstocks Materials Characterization.

47) James W Murray, Richard B Cook, Nicola Senin, Samer J Algodi and Adam T Clare, 2018. Defect-Free Tic/Si Multi-Layer Electrical Discharge Coatings Materials and Design.

48) Samer J. Algodi, James W. Murray, Adam T. Clare and Paul D. Brown, 2018. Modelling and Characterisation of Electrical Discharge Tic-Fe Cermet Coatings Procedia Cirp.

49) Alistair Speidel, Jonathon Mitchell-Smith, Ivan Bisterov And Adam T. Clare, 2018. The Dependence of Surface Finish On Material Precondition in Electrochemical Jet Machining Procedia Cirp.

50) Ivan Bisterov, Jonathon Mitchell-Smith, Alistair Speidel and Adam T. Clare, 2018. Specific and Programmable Surface Structuring by Electrochemical Jet Processing Procedia Cirp.

51) Jonathon Mitchell-Smith, Alistair Speidel, Ivan Bisterov and Adam T. Clare, 2018. Electrolyte Multiplexing in Electrochemical Jet Processing Procedia Cirp.

52) Ge Zhao, Ian Ashcroft, Richard Hague, Andrew Kennedy and Adam T. Clare, 2018. Salt-Metal Feedstocks for the Creation of Stochastic Cellular Structures with Controlled Relative Density by Powder Bed Fabrication Materials And Design.

53) Adam T. Clare, Alistair Speidel, Ivan Bisterov, Alexander Jackson-Crisp and Jonathon Mitchell-Smith, 2018. Precision Enhanced Electrochemical Jet Processing Cirp Annals.

54) Z Xu, C Hyde, L Sein, M Antar, J Dunleavey, R Cook And A.T Clare, 2018. The Influence of Shot Peening on the Fatigue Response of Ti-6al-4v Surfaces Subject to Different Machining Processes International Journal of Fatigue.

55) S.J. Algodi, J. Murray, P.D. Brown and A.T. Clare, 2018. Wear Performance of Tic/Fe Cermet Electrical Discharge Coatings Wear. 402-403, 109-123

56) Z. Xu, C.J. Hyde, C. Tuck And A.T. Clare, 2018. Creep Behaviour of Inconel 718 Processed by Laser Powder Bed Fusion Journal Of Materials Processing Technology.

57) J Murray, P B Brown, S Algodi And A T Clare, 2018. Formation Mechanism of Electrical Discharge Tic-Fe Composite Coatings Journal of Materials Processing Technology. [If – 2.359] (25%)

58) J. Mitchell-Smith, A. Speidel and A.T. Clare, 2018. Advancing Electrochemical Jet Methods Through Manipulation of the Angle of Address Journal of Materials Processing Technology. [If – 2.359] (40%)

59) J. Mitchell-Smith, A. Speidel and A.T. Clare, 2018. Transitory Electrochemical Masking for Precision Jet Processing Techniques Journal of Manufacturing Processes. [If – 2.322] (30%)

60) H Tan, W Shang, F. Zhang, A.T Clare, X. Lin, J. Chen and W. Huang, 2017. Process Mechanisms Based on Powder Flow Spatial Distribution in Direct Metal Deposition Journal of Materials Processing Technology. [If – 2.359] (20%)

61) H. Tan, W. Shang, K.S. Al-Hamdani, F. Zhang, Z. Xu and A.T. Clare, 2017. Direct Metal Deposition of Tib2/Alsi10mg Composites Using Satellite Powders Materials Letters. [If – 2.57] (40%)

62) M. Hirsch, P. Dryburgh, S. Catchpole-Smith, R. Patel, L. Parry, S. Sharples, I. Ashcroft and A.T. Clare, 2017. Targeted Rework Strategies for Powder Bed Additive Manufacture Additive Manufacturing. [Snip – 3.63 – No If] (30%)

63) O. Oyelola, P. Crawforth, R. M'saoubi And A.T Clare, 2017. On the Machinability of Directed Energy Deposited Ti6al4v Additive Manufacturing. [Snip – 3.63 – No If] (40%)

64) S. Algodi, P. Brown And A.T Clare, 2017. Modelling of Single Spark Interactions During Electrical Discharge Coating Journal of Materials Processing Technology 239 230-239 [If – 2.359] (50%)

65) F. Zhang, M. Mei, K. Al-Hamdani, H. Tan And A.T Clare, 2017. Novel Nucleation Mechanisms Through Satelliting in Direct Metal Deposition of Ti-15mo Materials Letters [If – 2.57] (40%)

66) M. Hirsch, S. Catchpole-Smith, R. Patel, P. Marrow, Wenqi Li, C. Tuck, S.D. Sharples And A.T. Clare, 2017. Meso-Scale Defect Evaluation of Selective Laser Melting Using Spatially Resolved Acoustic Spectroscopy Proceedings Of The Royal Society A [If – 2.450] (50%)

67) Xu Z, Thompson A, Leach R K, Hyde C, Maskery I, Tuck C And Clare A T, 2017. Staged Thermomechanical Testing of Nickel Superalloys Produced by Selective Laser Melting Materials And Design [If - 3.997] (40%)

68) F Zhang, M Yang, A.T. Clare, X. Lin, H Tan And Y Chen, 2017. Microstructure and Mechanical Properties of Ti-2al Alloyed With Mo Formed in Laser Additive Manufacture Journal of Alloys and Compounds. [If - 3.133] (30%)

69) D.L Bourell, J.P Kruth, M. Leu, G. Levy, D. Rosen, A. Beese and A.T. Clare, 2017. Materials For Additive Manufacturing Cirp Annals. [If – 4.1] (20%)

70) J Mitchell-Smith, A Speidel, J Gaskell and A.T. Clare, 2017. Energy Distribution Modulation by Mechanical Design for Electrochemical Jet Processing Techniques International Journal of Machine Tools and Manufacture. [If – 5.07] (30%)

71) A Colombi, V Ageeva, R Smith, A Clare, R Patel, M Clark, D Colquitt, P Roux, S Guenneau And R Craster, 2017. Enhanced Sensing and Conversion of Ultrasonic Rayleigh Waves by Elastic Metasurfaces Scientific Reports. [If – 4.85] (15%)

72) A.T Clare, A Speidel, J. Mitchell-Smith And S. Patwardhan, 2017. Electrolyte Design for Suspended Particulates in Electrolyte Jet Processing Cirp Annals. [If – 4.1] (60%)

73) J.C Walker, T.J Kamps, J. Mitchell-Smith and A.T. Clare, 2017. Tribological Behaviour of an Electrochemical Jet Machined Textured Al-Si Automotive Cylinder Liner Material Wear. [If – 2.8] (20%)

74) K.S. Al-Hamdani, J.W. Murray, A. Kennedy And A.T. Clare, 2017. Cold Sprayed Metal-Ceramic Coatings Using Satellite Powders Materials Letters [If – 2.57] (40%)

75) S. Catchpole-Smith, N. Aboulkhair, L. Parry, C. Tuck, I.A. Ashcroft And A.T. Clare, 2017. Fractal Scan Strategies for Selective Laser Melting of 'Unwieldable' Nickel Superalloys Additive Manufacturing. [Snip – 3.63 – No If] (25%)

76) Hirsch M, Patel R, Li W, Guan G, Leach R K, Sharples S D And Clare A T, 2017. Assessing the Capability of In-Situ Nondestructive Analysis During Layer Based Additive Manufacture Additive Manufacturing. 13, 135-142 [Snip – 3.63 – No If] (30%)

77) A Sumner, C Gerada, N Brown And A.T. Clare, 2017. Controlling DC Permeability in Cast Steels Journal of Magnetism and Magnetic Materials. [If – 2.63] (35%)

78) Samer J Algodi, James W Murray, Michael W Fay, Paul D Brown, Adam T Clare; 2017 Electrical Discharge Coating of Nanostructured Tic-Fe Cermets on 304 Stainless Steel Surface and Coatings Technology [If – 2.139] (20%)

79) Alistair Speidel, Adrian Hugh Alexander Lutey, Jonathon Mitchell-Smith, Graham A. Rance, Erica Liverani, Alessandro Ascari, Alessandro Fortunato, Adam Clare; 2017 Surface Modification of Mild Steel Using a Combination of Laser and Electrochemical Processes Surface and Coatings Technology [If – 2.139] (40%)

80) Catchpole-Smith, S; Clare, At; 2017 In-Situ Synthesis of Titanium Aluminides by Direct Metal Deposition Journal of Materials Processing Technology 239 230-239 [If – 2.359] (50%)

81) Guan, Guangying; Hirsch, Matthias; Syam, Wahyudin P; Leach, Richard K; Huang, Zhihong; Clare, Adam T; 2016 Loose Powder Detection and Surface Characterization in Selective Laser Sintering Via Optical Coherence Tomography Proceedings of The Royal Society A [If – 2.450] (50%)

82) Everton, Sarah K; Hirsch, Matthias; Stravroulakis, Petros; Leach, Richard K; Clare, Adam T; 2016 Review of In-Situ Process Monitoring and In-Situ Metrology for Metal Additive Manufacturing Materials & Design 95, 431-445 [If - 3.997] (30%)

83) Oyelola, Olusola; Crawforth, Peter; M'saoubi, Rachid; Clare, Adam T; 2016 Machining of Additively Manufactured Parts: Implications For Surface Integrity Procedia Cirp 45, 119-122 (40%)

84) Hirsch, Matthias; Dedenbach, Johannes; Lawes, Simon; Clare, Adam T; 2016 Focussed Arc Tungsten Inert Gas Brazing of Zinc-Coated Steels Proceedings of The Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture [If – 0.978] (60%)

85) Smith, Richard J; Hirsch, Matthias; Patel, Rikesh; Li, Wenqi; Clare, Adam T; Sharples, Steve D; 2016 Spatially Resolved Acoustic Spectroscopy for Selective Laser Melting Journal of Materials Processing Technology 236, 93-102 [If – 2.359] (30%)

86) Farayibi, PK; Abioye, TE; Clare, AT; 2016 A Parametric Study on Laser Cladding of Ti-6al-4v Wire and Wc/W2c Powder The International Journal of Advanced Manufacturing Technology [If – 1.556] (60%)

87) Guan, Guangying; Lu, Zeng H; Hirsch, Matthias; Goodridge, Ruth; Childs, David TD; Matcher, Stephen J; Clare, Adam T; Groom, Kristian M; 2016 Towards In-Situ Process Monitoring in Selective Laser Sintering Using Optical Coherence Tomography Spie Laser [If – 0.908] (60%)

88) Speidel, Alistair; Mitchell-Smith, Jonathon; Walsh, Darren A; Hirsch, Matthias; Clare, Adam; 2016 Electrolyte Jet Machining of Titanium Alloys Using Novel Electrolyte Solutions Procedia Cirp 42, 367-372 (60%)

89) Mitchell-Smith, J; Clare, AT; 2016 Electrochemical Jet Machining of Titanium: Overcoming Passivation Layers with Ultrasonic Assistance Procedia Cirp 42, 379-383 (60%)

90) Murray, James W; Clare, Adam T; 2016 Morphology and Wear Behaviour of Single and Multi-Layer Electrical Discharge Coatings Procedia Cirp 42, 236-239 (60%)

91) Abioye, TE; Farayibi, PK; Mccartney, DG; Clare, AT; 2016 Effect of Carbide Dissolution on the Corrosion Performance of Tungsten Carbide Reinforced Inconel 625 Wire Laser Coating Journal of Materials Processing Technology 231, 89-99 [If – 2.359] (60%)

92) Maskery, Ian; Aboulkhair, NT; Corfield, MR; Tuck, Christopher; Clare, AT; Leach, RK; Wildman, RD; Ashcroft, LA; Hague, RJM; 2016 Quantification and Characterisation of Porosity in Selectively Laser Melted Al–Si10–Mg Using X-Ray Computed Tomography Materials Characterization 111, 193-204 [If – 2.383] (15%)

93) Allwood, Julian M; Childs, Tom HC; Clare, Adam T; De Silva, Anjali Km; Dhokia, Vimal; Hutchings, Ian M; Leach, Richard K; Leal-Ayala, David R; Lowth, Stewart; Majewski, Candice E; 2016 Manufacturing at Double the Speed Journal of Materials Processing Technology 229, 729-757 [If – 2.359] (5%)

94) Murray, JW; Sun, J; Patil, DV; Wood, TA; Clare, AT; 2015 Physical and Electrical Characteristics of EDM Debris Journal of Materials Processing Technology 229, 54-60 [If – 2.359] (60%)

95) Algodi, SJ; Murray, JW; Clare, AT; Brown, Pd; 2015 Characterisation of Tic Layers Deposited Using an Electrical Discharge Coating Process Journal of Physics: Conference Series 644, 1 12008 [If – 0.54] (60%)

96) Guan, Guangying; Hirsch, Matthias; Lu, Zeng Hai; Childs, David TD; Matcher, Stephen J; Goodridge, Ruth; Groom, Kristian M; Clare, Adam T; 2015 Evaluation of Selective Laser Sintering Processes by Optical Coherence Tomography Materials & Design 88, 837-846 [If –] (60%)

97) Abioye, TE; Farayibi, PK; Kinnel, P; Clare, AT; 2015 Functionally Graded Ni-Ti Microstructures Synthesised in Process by Direct Laser Metal Deposition The International Journal of Advanced Manufacturing Technology [If – 1.779] (60%)

98) Farayibi, Pk; Abioye, Te; Murray, Jw; Kinnell, Pk; Clare, At; 2015 Surface Improvement Of Laser Clad Ti–6al–4v Using Plain Waterjet And Pulsed Electron Beam Irradiation Journal Of Materials Processing Technology [If – 2.359] (60%)

99) Abioye, TE; Mccartney, DG; Clare, AT; 2015 Laser Cladding of Inconel 625 Wire for Corrosion Protection Journal of Materials Processing Technology 217, 232-240 [If – 2.359] (60%)

100) Hirsch, Matthias; Dedenbach, Johannes; Lawes, Simon; Clare, Adam T; 2015 In Situ Low-Cost And Adaptable Braze Tool Evaluation System With Vision Analysis Proceedings Of The Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture 229, 9, 1595-1602 [If – 0.978] (60%)

101) Murray, JW; Walker, JC; Clare, AT; 2014 Nanostructures in Austenitic Steel After Edm and Pulsed Electron Beam Irradiation Surface And Coatings Technology 259, 465-472 [If - 2.139] (50%)

102) Li, Cl; Murray, JW; Voisey, KT; Clare, AT; Mccartney, DG; 2014 Corrosion Behaviour of a Rapidly Solidified Al 87.4 Co 7.9 Ce 4.7 Layer Prepared by Large Area Electron Beam Irradiation Applied Surface Science 320, 581-590 [If – 3.150] (25%)

103) Boud, F; Murray, JW; Loo, LF; Clare, AT; Kinnell, PK; 2014 Soluble Abrasives for Waterjet Machining Materials and Manufacturing Processes 1346-1352 [If – 1.63] (40%)

104) 川中拓磨; ムレージェームズ; クレアアダム; 国枝正典; 2013 電解液ジェットによるテクスチャリング 精密工学会学術講演会講演論文集 0, 979-980

105) Walker, JC; Murray, JW; Nie, Mengyan; Cook, RB; Clare, AT; 2014 The Effect of Large-Area Pulsed Electron Beam Melting on the Corrosion and Microstructure of a Ti6al4v Alloy Applied Surface Science 311, 534-540 [If – 3.150] (30%)

106) Kawanaka, Takuma; Kato, Shigeki; Kunieda, Masanori; Murray, James W; Clare, Adam T; 2014 Selective Surface Texturing Using Electrolyte Jet Machining Procedia Cirp 13, 345-349 (40%)

107) Presti, Davide Lo; Fecarotti, Claudia; Clare, Adam T; Airey, Gordon; 2014 Toward More Realistic Viscosity Measurements of Tyre Rubber–Bitumen Blends Construction and Building Materials 67,270-278 [If – 2.421] (30%)

108) Cl Li, JW Murray, KT Voisey, AT Clare, DG Mccartney 2014 Effect of Prior Laser Microstructural Refinement on the Formation of Amorphous Layer in an Al< Sub> 86</Sub> Co< Sub> 7.6</Sub> Ce< Sub> 6.4</Sub> Alloy Applied Surface Science 289, 230-236 [If – 2.538] (20%)

109) T Kawanaka, S Kato, M Kunieda, JW Murray, AT Clare 2014 Selective Surface Texturing Using Electrolyte Jet Machining Procedia Cirp 13, 345-349 (20%)

110) F Boud, JW Murray, LF Loo, AT Clare, PK Kinnell 2014 Soluble Abrasives for Waterjet Machining Materials and Manufacturing Processes [If – 1.486] (40%)

111) JC Walker, JW Murray, M Nie, RB Cook, AT Clare 2014 The Effect of Large-Area Pulsed Electron Beam Melting on the Corrosion and Microstructure of a Ti6al4v Alloy Applied Surface Science [If – 2.538] (40%)

112) D Lo Presti, C Fecarotti, At Clare, G Airey 2014 Toward More Realistic Viscosity Measurements of Tyre Rubber–Bitumen Blends Construction and Building Materials [If - 2.265] (20%)

113) M Hirsch, J Dedenbach, S Lawes, At Clare 2014 In Situ Low-Cost and Adaptable Braze Tool Evaluation System with Vision Analysis Proceedings of the Institution of Mechanical Engineers, Part B: Journal of Engineering Manufacture [If - 0.661] (40%)

114) PK Farayibi, JW Murray, L Huang, F Boud, PK Kinnell, AT Clare* 2014 Erosion Resistance Of Laser Clad Ti-6al-4v/Wc Composite for Waterjet Tooling Journal of Materials Processing Technology 214 (3), 710-721 [If - 2.041] (50%)

115) A.Okada, Y. Okamoto, Y. Uno And A.T. Clare* 2013. Effect Of EB Polishing on Releasability of Molded Rubber from Metal Mold Tools International Journal Of Mechanics And Materials In Design [If - 0.732] (20%)

116) T. E. Abioye, J. Folkes And A. T. Clare 2013 A Parametric Study of Inconel 625 Wire Laser Deposition Journal of Materials Processing Technology [If - 2.041] (50%)

117) C.L. Li, J.W. Murray, K.T. Voisey, A.T. Clare And D.G. Mccartney 2013. Amorphous Layer Formation in an Al-Co-Ce Glass-Forming Alloy by Large-Area Electron Beam Irradiation Applied Surface Science. [If – 2.538] (20%)

118) Walker, J.C., Cook, R.B., Murray, J.W. And Clare, A.T., 2013. Pulsed Electron Beam Surface Melting of Cocrmo Alloy For Biomedical Applications Wear. [If – 1.862] (40%)

119) J. W. Murray, M. W. Fay, M. Kunieda And A. T. Clare, 2013. Tem Study on the Electrical Discharge Machined Surface of Single-Crystal Silicon Journal of Materials Processing Technology. [If - 2.041] (50%)

120) P.K. Farayibi, T.E. Abioye, And A.T. Clare, 2013. Abrasive Wear Performance Of Laser Deposited Ti-6al-4v Wire/Wc Powder Surface Engineering [If – 1.51] (50%)

121) T. E. Abioye, J. Folkes, A. T. Clare And D. G. Mccartney, 2013. Concurrent Inconel 625 Wire and WC Powder Laser Cladding: Process Stability and Microstructural Characterisation Surface Engineering. [If – 1.51] (40%)

122) J. Murray, P. Kinnell, A. Cannon, B. Bailey And A.T. Clare*, 2013 Surface Finishing of Intricate Metal Mold Structures by Large-Area Electron Beam Irradiation Precision Engineering. [If – 1.403] (50%)

123) A.T. Clare*, O. Oyelola, P. Farayibi And J. Folkes, 2012. Laser Cladding for Railway Repair and Preventative Maintenance Journal Of Laser Applications. [If - 1.338] (30%)

124) J. Murray, D. Zdebski And A.T. Clare*, 2012. Workpiece Debris Deposition on Tool Electrodes and Secondary Discharge Phenomena in Micro-Edm Journal of Materials Processing Technology. [If - 2.041] (50%)

125) J. Murray And A.T. Clare*,2012. Repair of EDM Induced Surface Cracks by Pulsed Electron Beam Irradiation Journal Of Materials Processing Technology. [If - 2.041] (50%)

126) P. K. Farayibi, J. A. Folkes And A. T. Clare*, 2012. Laser Deposition of Ti-6al-4v Wire with WC Powder for Functionally Graded Components Materials and Manufacturing Processes. [If – 1.486] (50%)

127) A.T. Clare*, O. Oyelola, T.E. Abioye And P. K. Farayibi, 2012 Laser Cladding of Rail Steel With Co-Cr Surface Engineering. [If – 1.51] (50%)

128) P.K. Farayibi, J. Folkes, A.T. Clare And O. Oyelola, 2011.Cladding Of Pre-Blended Ti-6al-4v and WC Powder for Wear Resistant Applications Surface & Coatings Technology. [If – 2.199] (50%)

129) J. C. Walker, J. Murray, S. Narania And A. T. Clare, 2011.Dry Sliding Friction and Wear Behaviour of an Electron Beam Melted Hypereutectic Al–Si Alloy Tribology Letters. [If – 2.151] (50%)

130) A.T. Clare*, L. Gao, B. Brkic, P.R. Chalker And S. Taylor, 2010.Linear Ion Trap Fabricated Using Rapid Manufacturing Technology Journal of the American Society for Mass Spectrometry. [If – 3.193] (50%)

131) B. Brkic, N. France, A.T. Clare, C.J. Sutcliffe, P.R. Chalker and S. Taylor,2009. Development of Quadrupole Mass Spectrometers Using Rapid Prototyping Technology Journal of the American Society for Mass Spectrometry. [If – 3.193] (50%)

132) A.T. Clare, P.R. Chalker, S. Davies, Cj Sutcliffe and S. Tsopanos, 2008 Selective Laser Sintering of Barium Titanate-Polymer Composite Films Journal of Materials Science. [If – 2.041] (50%)

133) A.T. Clare, P.R. Chalker, S. Davies, S. Tsopanos, And C.J. Sutcliffe, 2008.Selective Laser Melting of High Aspect Ratio 3d Nickel–Titanium Structures Two Way Trained for Mems Applications International Journal of Mechanics and Materials in Design [If – 0.732] (60%)

134) P.R. Chalker, A.T. Clare, S. Tsopanos, C.J. Sutcliffe, And S. Davies, 2006. Selective Laser Melting of High Aspect Ratio 3d Nickel-Titanium Structures for Mems Applications Materials Research Society Proceedings (50%)

Selected Conference Papers (All Peer Reviewed)

1) J.W. Murray, A.T. Clare, Improved Surface Properties Of Edm Components After Irradiation by Pulsed Electrons. Matador Conference Manchester, 2012.

2) N. Vladov, J.W Murray, A.T. Clare, J. Segal, S. Ratchev Surface-Finishing of Focused Ion Beam Milled Features by Pulsed Electron Irradiation. Eipbn Conference 2012.

3) C.L. Li, J.W Murray, K.T. Voisey, A. T. Clare, D. G. Mccartney. Amorphous Layer Formation in an Al-Co-Ce Alloy Using Electron Beam Melting and Laser Microstructural Refinement. SMT Conference 2012.

4) J.C. Walker, R.B. Cook, J.W Murray, A.T. Clare. Pulsed Electron Beam Surface Melting of Cocrmo Alloy for Biomedical Applications. Wear of Materials Conference 2013.

5) T. Kawanaka, S. Kato, M. Kunieda, J.W. Murray, A.T. Clare Micro Textured Surfaces Created by Electrolyte Jet Machining. Aspen Conference 2013 (Under Review)

6) Chalker, Pr, Clare, At, Tsopanos, S, Sutcliffe, Cj And Davies, S, 2006. Selective Laser Melting of High Aspect Ratio 3D Nickel-Titanium Structures for Mems Applications Materials Research Society Symposium Proceeding. 890, 93-98

7) C. Hauser, Clare, AT, D.M. Lewis, P.J. Broadbent and C.J. Sutcliffe,2009.Polymer Ink Development and Adaptive Slicing Techniques in Three Dimensional Printing in: International Solid Freeform Fabrication Symposium.

8) Clare, AT, Chalker, PR, Jetinder Singh, C. Hauser And Sutcliffe, CJ, 2009. Rapid Prototyping for Chemical Reactor and Chemical Analysis Applications In: International Solid Freeform Fabrication Symposium.

9) Jetinder Singh, C. Hauser, A.T. Clare, P.R. Chalker And C.J. Sutcliffe, 2009. Customized Layer Deposition for Rapid Prototyping *In:* International Solid Freeform Fabrication Symposium.

10) Clare, AT, Folkes, J, Farayibi, P, Segal J And Oyelola O,2011.Laser Deposition from White Metals to Hard Facing Materials (Sic) *In:*Industrial Laser Applications Symposium (Ilas).

11) Farayibi, P, Folkes, J And Clare, At,2011. Influence of Process Parameters and Energy Density on the Microstructure Of Ti-6al-4v Wire/Wc Powder Cladding *In:* The International Congress on Applications of Lasers & Electro-Optics (Icaleo),.

12) 2011. Laser Cladding for Railway Repair and Preventative Maintenance *In:* The International Congress on Applications of Lasers & Electro-Optics (Icaleo). 240-249

13) P. K. Farayibi, T.E. Abioye And A.T. Clare,2012.Microstructural Characterisation and Wear Performance of Laser Deposited Ti-6al-4v Wire/Wc Powder *In:* 26th International Conference on Surface Modification Technologies at Lyon, France

14) Maculotti G, Senin N, Oyelola O, Galetto M, Clare A And Leach R K, 2019. Multi-Sensor Data Fusion for the Characterisation of Laser Cladded Cermet Coatings *In:* Proc. 19th Int. Euspen Conf., Bilboa, Spain, Jun.

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